

```
DATABASE_PASSWORD
RABBIT_PASSWORD
SERVICE_TOKEN
SERVICE_PASSWORD
ADMIN_PASSWORD
```

If we specify the option `OFFLINE=True` in the `localrc` file inside DevStack directory, and if after specifying this, we run `stack.sh`, it will not check any parameter over the Internet. It will set up DevStack using all the packages and code residing in the local system. In the phase of code development, there is need to commit the local changes in the `/opt/stack/nova` repository before `restack` (re-running `stack.sh`) with the `RECLONE=yes` option. Otherwise, the changes will not be committed.

To use more than one interface, there is a need to specify which one to use for the external IP using this configuration:

```
HOST_IP=xxx.xxx.xxx.xxx
```

Cinder on DevStack

Cinder is a block storage service for OpenStack that is designed to allow the use of a reference implementation (LVM) to present storage resources to end users that can be consumed by the OpenStack Compute Project (Nova). Cinder is used to virtualise the pools of block storage devices. It delivers end users with a self-service API to request and use the resources, without requiring any specific complex knowledge of the location and configuration of the storage where it is actually deployed.

All the Cinder operations can be performed via any of the following:

1. CLI (Cinder's `python-cinderclient` command line module)
2. GUI (Using OpenStack's GUI project `horizon`)
3. Direct calling of Cinder APIs

Creation and deletion of volumes: To create a 1 GB Cinder volume with no name, run the following command:

```
$ cinder create 1
```

To see more information about the command, just type `cinder help <command>`

```
$ cinder help create
```

```
usage: cinder create [--snapshot-id <snapshot-id>]
                  [--source-uuid <source-uuid>]
                  [--image-id <image-id>]
                  [--display-name <display-name>]
                  [--display-description <display-
description>]
                  [--volume-type <volume-type>]
                  [--availability-zone <availability-
zone>]
                  [--metadata [<key=value> [<key=value>
```

```
...]]]
                                <size>
Add a new volume.
Positional arguments:
  <size>                        Size of volume in GB
Optional arguments:
  --snapshot-id <snapshot-id>  Create volume from snapshot id
                                (Optional,
                                Default=None)
  --source-uuid <source-uuid>  Create volume from volume id
                                (Optional, Default=None)
  --image-id <image-id>        Create volume from image id
                                (Optional, Default=None)
  --display-name <display-name> Volume name (Optional, Default=None)
  --display-description <display-description> Volume description (Optional,
                                Default=None)
  --volume-type <volume-type>  Volume type (Optional, Default=None)
  --availability-zone <availability-zone> Availability zone for volume
                                (Optional, Default=None)
  --metadata [<key=value> [<key=value> ...]] Metadata key=value pairs (Optional,
                                Default=None)
```

To create a Cinder volume of size 1GB with a name, using `cinder create --display-name myvolume`:

```
$ cinder create --display-name myvolume 1
+-----+-----+
| Property | Value |
+-----+-----+
| attachments | [] |
| availability_zone | nova |
| bootable | false |
| created_at | time |
| display_description | None |
| display_name | myvolume |
| id | id |
| metadata | {} |
| size | 1 |
| snapshot_id | None |
| source_uuid | None |
| status | creating |
| volume_type | None |
+-----+-----+
```

To list all the Cinder volumes, using `cinder list`:

```
$ cinder list
```